

Klever Klues

Platform Architecture & Developer Documentation

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Pre-Development Specification, Project Structure & Research Insights

1. Executive Summary & Problem Statement

Klever Klues is an integrated digital healthcare marketplace and Mental Wellbeing & Prevention (MWP) platform. Mental health issues (stress, anxiety, depression) are rising, and the platform addresses the absence of simple, preventive, and structured solutions that enable early identification of mental health concerns.

Critical Mission: A core pillar of Klever Klues is its life-saving **Emergency SOS System**, built to utilize zero-latency geofenced real-time broadcasting to bridge patients in crisis with verified local volunteers and emergency services instantly.

2. Platform Architecture & Features

The system is designed with a multi-faceted architecture supporting different user roles and features based on extensive market research.

2.1 User Modules & Flow

- **User Access:** Registration/login, language selection, and an **Anonymous Mode** option to protect identities during consultations.
- **Consultation Flow:** Selection -> Secure Payment -> Online Consultation (Chat/Audio/Video) -> Digital Prescription Generation.
- **Vendor/Consultant System:** Registration, document submission, and a strict verification workflow (Pending -> Approved/Rejected/Suspended).
- **Admin Dashboard:** Centralized hub for application review, feedback provision, and vendor status governance.

2.2 Key Features

- **Doctor & Hospital Discovery:** Search by specialization, location, and ratings. View detailed profiles and book appointments.
- **E-Pharmacy:** Browse medicines categorized by mental wellness (Anxiety, Depression, Sleep, etc.) and upload prescriptions.
- **Health & Wellbeing Blog:** Educational content categorized by Mental Health, Wellbeing, Lifestyle, and Self-Care.

- **Emergency SOS System:** Dedicated activation button integrating with background-verified volunteers and location-based dispatch.

3. Technology Stack

To ensure high performance, real-time emergency capabilities, and strict healthcare compliance (End-to-End Encryption, No Third-Party API Integration for core chat), the platform utilizes a decoupled full-stack architecture.

3.1 Frontend Architecture

- **Framework:** React 18 with Vite.
- **Routing:** React Router v6.
- **Styling:** Tailwind CSS v4 configured with custom design tokens.
- **State Management:** Zustand (for Auth, Cart, and SOS).
- **Data Fetching:** @tanstack/react-query.
- **UI/Animations:** Framer Motion and Lucide React.

3.2 Backend & Infrastructure

- **Runtime & Real-time:** Node.js paired with WebSockets for the sub-second SOS dispatch system.
- **Database:** Supabase (PostgreSQL).
- **Geospatial Logic:** PostGIS extension for complex radius-matching calculations.

4. Design System & Theming

The UI is governed by the "Serene Assurance" palette, configured directly within `tailwind.config.js`.

Token	Color Value	Usage
Primary	#00685c (Seafoam Teal)	Primary buttons, active states, major headings.
Secondary	#3d665e (Muted Sage)	Secondary buttons, subtle accents, secondary text.
Tertiary	#864f13 (Soft Peach/Warm Brown)	Warning states, highlights, warm accents.
Background	#f5fbf8 (Mint Tint)	Global application background, ensuring reduced eye strain.

4.1 Foundational UI Components

- **Buttons:** Unified `Button.jsx` supporting primary, secondary, and ghost variants.
- **Cards:** Standardized `Card.jsx` containers for Doctor Profiles, Hospitals, and Pharmacy items.
- **Forms & Inputs:** Fully controlled components utilizing standard shadcn-style inputs.
- **SOS Component:** High-contrast state and pulse animation for immediate visibility.

5. Directory & Project Structure

The codebase follows a highly modular, feature-based architecture.

```

klever-klues/
├─ public/
├─ src/
│   ├─ assets/                # Logo, icons, images
│   ├─ components/
│   │   ├─ ui/                # Base UI components (shadcn pattern)
│   │   ├─ shared/            # Navbar, Footer, SOSButton, Cards
│   │   └─ forms/              # LoginForm, RegisterForm
│   ├─ layouts/
│   │   ├─ RootLayout.jsx      # Global wrapper (Navbar + Footer)
│   │   ├─ AuthLayout.jsx      # Centered layout for authentication
│   │   └─ DashboardLayout.jsx # Sidebar layout for user/admin portals
│   ├─ pages/
│   │   ├─ marketplace/        # Public Medical Marketplace
│   │   │   └─ Home, Doctors, Hospitals, Pharmacy, Blog
│   │   ├─ mwp/                # Mental Wellbeing & Prevention
│   │   │   └─ Landing, Consultants, Resources
│   │   ├─ auth/                # Authentication flows
│   │   │   └─ RoleSelect, Login, Register
│   │   └─ dashboard/           # Protected Portals
│   │       └─ UserDashboard, ConsultantDashboard, AdminDashboard
│   ├─ store/                  # Zustand state management
│   │   ├─ authStore.js
│   │   ├─ cartStore.js
│   │   └─ sosStore.js
│   ├─ hooks/                  # Custom business logic hooks
│   │   └─ useAuth.js, useCart.js, useSOS.js
│   ├─ lib/                    # Utilities and theme constants
│   ├─ router/
│   │   └─ index.jsx            # React Router configuration
│   ├─ App.jsx
│   ├─ main.jsx
│   └─ index.css
├─ tailwind.config.js          # Design tokens
└─ package.json

```

6. Competitor & Market Analysis Integration

Research indicates a significant treatment gap (70-92% in India) due to stigma and a shortage of professionals. Klever Klues is positioned to address these gaps through technology.

- **Government Initiatives:** Offer large-scale helplines (e.g., Tele MANAS) but face uneven implementation and digital divide challenges. *Klever Klues bridges this with a highly optimized, accessible web application.*
- **NGOs:** Provide strong community outreach but face funding limitations. *Klever Klues implements a sustainable marketplace model to ensure continuous service availability.*
- **Corporate Sector:** Focuses on employee assistance but lacks long-term monitoring. *Klever Klues provides detailed dashboards for ongoing user health tracking.*
- **Global Apps (Wysa, Headspace):** Often rely on AI chatbots for mild stress. *Klever Klues differentiates by providing direct access to licensed professionals and a dedicated SOS crisis system.*

7. Development Implementation Roadmap

The build process is structured into 5 iterative phases.

Phase 1: Foundation (Completed)

Vite + React initialization. Tailwind configuration. React Router setup. Global layouts established.

Phase 2: Authentication Flow

Implementation of the RoleSelect, Login, and Register views. Integration of `authStore.js` using Zustand.

Phase 3: Marketplace Ecosystem

Building Home page, Doctor/Hospital listings with filters, detailed profiles with booking, and E-Pharmacy with cart capabilities.

Phase 4: Mental Wellbeing & Dashboards

Constructing MWP landing page, Consultant directories, secure messaging, and role-specific dashboards (Admin verification, User tracking).

Phase 5: Real-Time SOS & Polish

Finalizing `SOSButton.jsx` integration with `useSOS.js` for geolocation.
Implementing animations, error boundaries, and accessibility audits.